

Neural Holographic Near-eye Displays for Virtual Reality

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Figure 1: Experimental holograms produced by our prototype near-eye display using Neural Holography algorithms. Our synthesized holograms accurately generate high quality 3D focus cues. These images of our holograms are captured by a camera positioned at the eyebox of our display. The camera is focused far on the background in the left column and close on the foreground in the right column. Source images by © Blender Foundation.

ABSTRACT

By manipulating light as a wavefront, holographic displays have the potential to revolutionize virtual reality (VR) and augmented reality (AR) systems. These displays support 3D focus cues for visual comfort, vision correcting capabilities, and high light efficiency. However, despite their incredible promise, holographic displays have consistently been hampered by poor image quality. Recently, artificial intelligence-driven computer-generated holography (CGH)

algorithms have emerged as a solution to this obstacle. On a prototype holographic display, we demonstrate how the progress of recent state-of-the-art Neural Holography algorithms can produce high-quality dynamic 3D holograms with accurate focus cues. The advances demonstrated in this work aim to provide a glimpse into a future where our displays can fully reproduce three-dimensional virtual content.

CCS CONCEPTS

• Hardware → Emerging technologies; • Computing methodologies → Computer graphics.

KEYWORDS

computational displays, holography, augmented reality, virtual reality

ACM Reference Format:

Suyeon Choi, Manu Gopakumar, Brian Chao, Gun-Yeal Lee, Jonghyun Kim, and Gordon Wetzstein. 2023. Neural Holographic Near-eye Displays for

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SIGGRAPH '23 Emerging Technologies, August 06–10, 2023, Los Angeles, CA, USA
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ACM ISBN 979-8-4007-0154-2/23/08.
<https://doi.org/10.1145/3588037.3595395>

Virtual Reality. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH '23 Emerging Technologies)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588037.3595395>

1 INTRODUCTION

Emerging virtual reality (VR) and augmented reality (AR) near-eye displays will enable unprecedented experiences across entertainment, education, communication, training, and many other domains. For these applications, near-eye displays will be pushed to satisfy a challenging set of criteria for compactness, light efficiency, and perceptual realism. Holographic displays have the potential to go far beyond the capabilities of conventional 2D displays towards satisfying these requirements. By shaping emitted wavefronts of light, computer-generated holography (CGH) can reproduce full 3D depth cues, correct for visual aberrations of the user or downstream optics, and offer high brightness with low power consumption. However, these displays have not yet achieved widespread usage because of long-standing limitations such as computational efficiency, compactness, and image quality.

Artificial intelligence-driven CGH algorithms have presented solutions to many of these issues. Neural network-based hologram synthesis can now generate 2D and 3D holograms in real-time [Peng et al. 2020; Shi et al. 2021, 2022], and the co-design of algorithms and optical systems has led to VR holographic displays in a glasses-like form factor [Gopakumar et al. 2021; Kim et al. 2022]. In this work, we demonstrate the culmination of recently developed Neural Holography algorithms that have steadily improved the image quality of holographic displays [Choi et al. 2022, 2021; Peng et al. 2020]. Conventionally, holographic displays have suffered from speckle and coherent ringing artifacts which would not be acceptable in a consumer display. These aberrations primarily result due to a mismatch between simulated wave propagation models and the actual propagation through holographic displays. Neural Holography techniques use learned wave propagation models to very accurately simulate the propagation through holographic displays. By overcoming this *sim2real* mismatch, these models enable the synthesis of very high quality holograms that are free from the aberrations of typical holographic displays.

2 METHODS

Most holographic displays use a phase-only spatial light modulator (SLM) to control the phase of a wavefront from a coherent light source, i.e., a laser. The resulting diffracted wavefront propagates to produce a desired 3D intensity distribution. Typically, CGH algorithms use simulated propagation models, such as the angular spectrum method (ASM), to compute diffraction patterns that produce a variety of desired intensity distributions including 2D images, 3D focal stacks, and 4D light fields. ASM is a plane-to-plane propagation model derived from the scalar wave equation. Unfortunately, ASM propagation does not match the physical propagation well, since it does not account for the non-idealities of the display components. As a result, the physically recreated intensity distributions are usually heavily aberrated when SLM diffraction patterns are computed with ASM propagation. Here, we demonstrate how learned propagation models can enable simulated propagation to

match the physical propagation and produce high-quality experimental holograms.

The learned propagation models that we use here are the culmination of several recent Neural Holography publications [Choi et al. 2022, 2021; Peng et al. 2020]. The core idea behind this family of techniques is to improve the accuracy of simulated wave propagation models by integrating physically-inspired learned parameters and convolutional neural networks (CNNs) applied to complex-valued wavefronts during the propagation. With these more accurate models, we have been able to synthesize holograms with substantially improved image quality. Most recently, by pairing our learned propagation models with a differentiable quantization layer, we constructed a framework to produce state-of-the-art 3D holograms with a natural defocus behavior using fast, heavily-quantized SLMs [Choi et al. 2022].

3 EXPERIENCE

In this installation, we will provide attendees with the opportunity to directly view the dynamic high quality 3D holograms from our display. We will synchronize our eye-safe laser illumination with our high speed SLM to display field-sequential full color content. Our setup will have a beamsplitter that splits the output of our display into two beam paths. On one beam path, we will position a camera that is focusing in and out of the output of our display. A live feed of the hologram as seen by this camera will be shown on a monitor in our booth. Examples of the views that can be seen by a camera through our setup are shown in Figure 1. On the second beam path, we will allow attendees to directly view the output of this display with their eyes. Crucially, this will showcase the perceptual benefits of the algorithmic innovations that we have developed. To the best of our knowledge, this will be the first live demonstration of high quality 3D content from a holographic display. In 2020, there was an emerging technologies demonstration with early Neural Holography techniques. However, due to COVID-19, attendees did not have an opportunity to see the holograms live. Since then, our methods have significantly improved, and we're ready to show the improved results to attendees now.

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